

Nested if-else

Example (1):

Write a C++ program to Check the Greatest Among Three Numbers demonstrate the use of nested if-else statements.

Sol.

```
#include <iostream>
using namespace std;
int main() {
    int a = 10, b = 12, c = 16;
    // Outermost if else
    if (a < b) {
        // Nested if else
        if (c < b) {
            cout << "the largest is b with value: " << b;
        }
        else {
            cout << "the largest is c with value:" << c;
        }
    }
    else {
        // Nested if else
        if (c < a) {
            cout << "the largest is a with value: " << a;
        }
        else {
            cout << "the largest is c with value: " << c;
        }
    }
    return 0;
}
```

The output:

The largest is c with value: 16

Switch selection statement

Example (1):

Write a C++ program to read two integer numbers, and select the operation to perform on these numbers:

```
#include <iostream>
#include <stdlib.h>
using namespace std;
int main()
{
    // Switch variable for operation choice
    char c;
    // Operands
    int x, y;
    cout << "Enter the two numbers: ";
    cin >> x >> y;
    cout << "Enter the Operator (+,-,*,/)\nEnter any other to "
        "exit\n";
    cin >> c;
    // switch case with operation for each operator
    switch (c)
    {
        case '+':
            cout << x << " + " << y << " = " << x + y << endl;
            break;
        case '-':
            cout << x << " - " << y << " = " << x - y << endl;
            break;
        case '*':
            cout << x << " * " << y << " = " << x * y << endl;
            break;
        case '/':
            cout << x << " / " << y << " = " << x / y << endl;
            break;
        default:
            printf("Exiting\n");
    }

    return 0;
}
```

Output:

Enter the two numbers: 5 4

Enter the Operator (+,-,*,/)

Enter any other to exit

+

5 + 4 = 9

Laboratory task to groups A, B, C:

Develop the program using switch statement to display the estimation of degrees for one student based on the range of degree in one subject. By benefited from the following table.

The degree	The estimation
≥ 90	Excellent
$\geq 80 \ \& \ < 90$	Very good
$\geq 70 \ \& \ < 80$	good
$\geq 60 \ \& \ < 70$	accepted
< 60	Fail